

- Lab 9 : Data Analytics
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Lab 10 – Inference for Categorical Data

Exercise 1:

Some people claim that they can tell the difference between a diet soda and a regular soda in the first sip. A researcher wanting to test this claim randomly sampled 80 such people. He then filled 80 plain white cups with soda, half diet and half regular through random assignment, and asked each person to take one sip from their cup and identify the soda as diet or regular. 53 participants correctly identified the soda.

- Do these data provide strong evidence that these people are any better or worse than random guessing at telling the difference between diet and regular soda?
- Interpret the p-value in this context.

Exercise 2:

A group of friends debated whether more men use smartphones than women. They consulted a research study of smartphone use among adults. The results of the survey indicate that of the 973 men randomly sampled, 379 use smartphones. For women, 404 of the 1304 who were randomly sampled use smartphones.

- Is there evidence that the proportion of men who use smartphones is higher than that of women who use smartphones? Formulate the appropriate hypotheses and test at the 5% level of significance.
- Determine the 95% confidence interval for the difference between the two proportions.

Exercise 3:

Researchers studying the link between prenatal vitamin use and autism surveyed the mothers of a random sample of children aged 24 - 60 months with autism and conducted another separate random sample for children with typical development. The table below shows the number of mothers in each group who did and did not use prenatal vitamins during the three months before pregnancy (periconceptional period).

		Autism		Total
		Autism	Typical development	
<i>Periconceptional prenatal vitamin</i>	No vitamin	111	70	181
	Vitamin	143	159	302
	Total	254	229	483

- State appropriate hypotheses to test for independence of use of prenatal vitamins during the three months before pregnancy and autism.
- Complete the hypothesis test and state an appropriate conclusion. (Reminder: Verify any necessary conditions for the test.)
- A New York Times article reporting on this study was titled "Prenatal Vitamins May Ward Off Autism". Do you find the title of this article to be appropriate? Explain your answer. Additionally, propose an alternative title.

End of the lab.